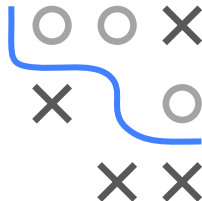


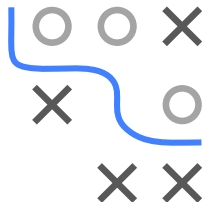
BATCH LEARNING

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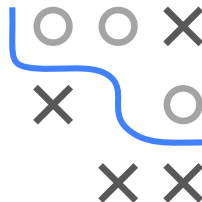
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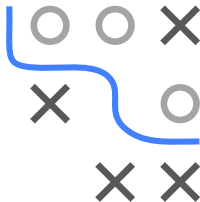
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- The learning task on the available data beforehand is called the *training phase* and the prediction on the unseen data is called the *testing phase*. Both phases are **separated**.

Batch Learning



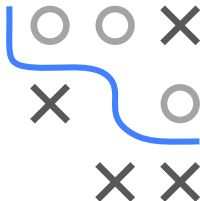
ONLINE LEARNING

- However, many real-world problems are *dynamic* with the following aspects:
 - *Sequential order* — data is generated only bit by bit;
 - *On-the-fly decisions* — decisions or predictions have to be made during the data generating process;
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ONLINE LEARNING

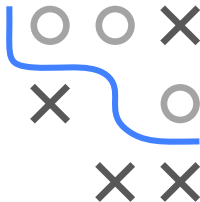
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- ```
graph TD; subgraph "Online Learning"; Apply([Apply]) --> GetMoreData([Get More Data]); GetMoreData --> Analyse([Analyse]); Analyse --> Apply; end
```
- The diagram illustrates the Online Learning cycle. It consists of three blue oval nodes arranged in a triangle: 'Apply' at the top, 'Analyse' at the bottom left, and 'Get More Data' at the bottom right. Yellow curved arrows connect them in a clockwise cycle: from 'Apply' to 'Get More Data', from 'Get More Data' to 'Analyse', and from 'Analyse' back to 'Apply'. The entire cycle is enclosed in a dashed-line box labeled 'Online Learning' at the top.
- These dynamic aspects outline the framework where **online learning** is settled.
  - Characteristically: In the online learning scenario the training phase and the testing phase are **interleaved**.



# ONLINE LEARNING: EXAMPLES

There are many real-world applications which fit into the online learning scenario:

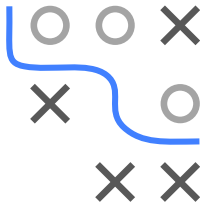
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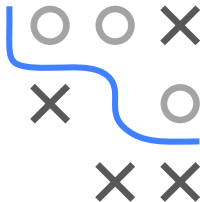
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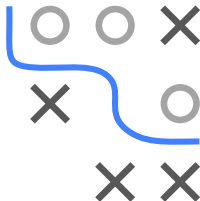
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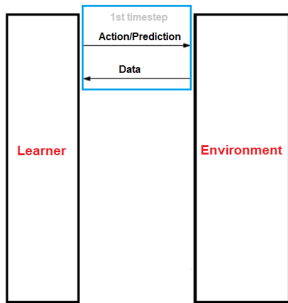
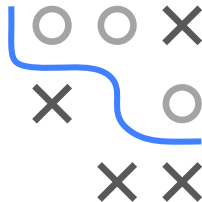
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- *Navigation systems* — Find the best path from A to B given the current traffic situation.
- *Autonomous driving systems* — Steer the automotive, while constantly monitoring the environment and react appropriately to any changes.
- ...



# ONLINE LEARNING: ILLUSTRATION

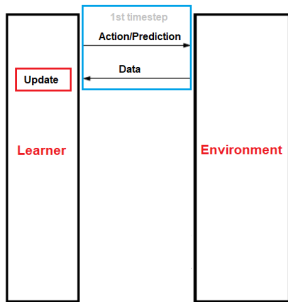
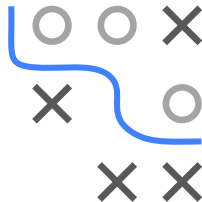
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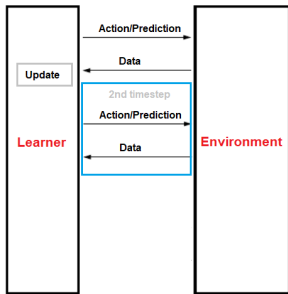
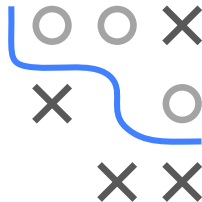
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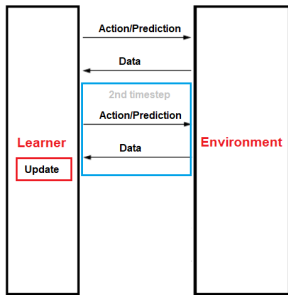
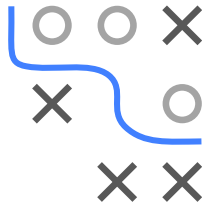
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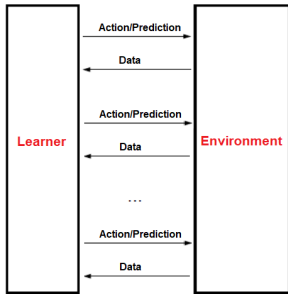
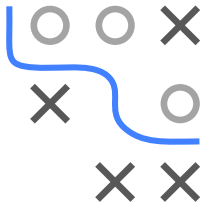
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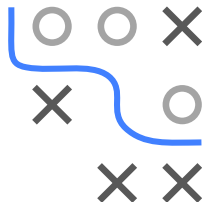


⇒ The learner and the environment are alternately performing their actions.

# THE BASIC ONLINE LEARNING PROTOCOL

Formally, an online learning problem consists of:

- a learner (forecaster, agent resp. decision maker), an environment (user resp. adversary, system resp. nature),
- time steps  $1, 2, \dots, T$  (may be infinite),
- available actions  $\mathcal{A}$  for the learner (may be infinite),
- environmental data space  $\mathcal{Z}$ ,
- a loss function  $L : \mathcal{A} \times \mathcal{Z} \rightarrow \mathbb{R}$ .



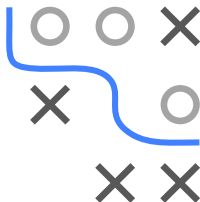
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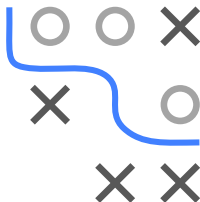
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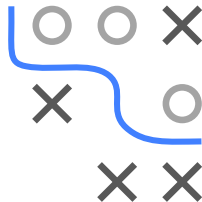
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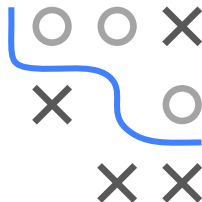
# THE EXTENDED ONLINE LEARNING PROTOCOL

- In some applications, the environmental data consists of two parts:  
 $z_t = (z_t^{(1)}, z_t^{(2)})$ , where the first part of the data,  $z_t^{(1)}$ , is revealed to the learner **before** the action is made. After the learner carries out its action, the remaining part of the environmental data is revealed, that is,  $z_t^{(2)}$ .



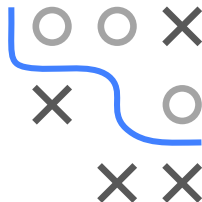
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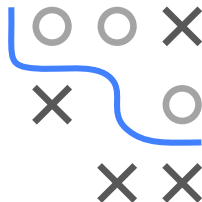
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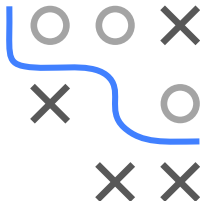
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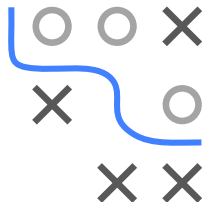
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- Typically  $\mathcal{A} = \mathcal{Y}$  and  $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ , so that
  - the first part  $z_t^{(1)} = \mathbf{x}_t$  is some feature information,
  - the learner's chosen action  $a_t = \hat{y}_t$  corresponds to a prediction (dep. on  $\mathbf{x}_t$ ),
  - the second part  $z_t^{(2)} = y_t$  is the corresponding outcome.



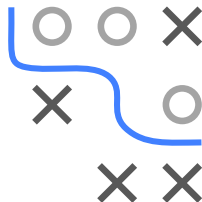
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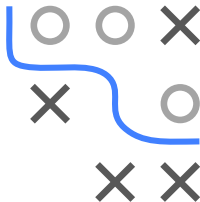
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- In particular, the environmental data are not necessarily generated by a probability distribution!
- This also covers the area of *adversarial learning*: the data can even be generated by an adversary trying to fool the learner.
- However, the online learning setting can of course also be considered in a statistical setting.





# ONLINE LEARNING: REQUIREMENTS

- The dynamical aspects have to be incorporated for the design of efficient learning algorithms.
- The online learner has to cope with the sequential availability of the data and to cope with time as well as computational constraints.
- Roughly speaking, one seeks to construct an online learning algorithm which is adaptive to the environment and allows incremental as well as preferably cheap updates over time.
- Although consideration of time and memory constraints is important for practical purposes, we will only implicitly consider these constraints in this lecture.
- We will mainly focus our theoretical analysis on the performance of the learner in terms of its (cumulative) loss, which, however, will usually ignore computational aspects of the learner.

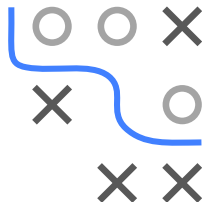


# MEASURE OF QUALITY IN ONLINE LEARNING

- In order to measure the quality of an online learner one can compute the difference between the cumulative loss of the learner and the cumulative loss by taking some competing action  $a \in \mathcal{A}$  :

$$R_T(a) = \sum_{t=1}^T L(a_t, z_t) - \sum_{t=1}^T L(a, z_t).$$

- This value is called the *(cumulative) regret of a learner* with respect to an action  $a \in \mathcal{A}$ .

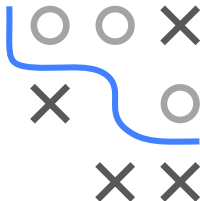


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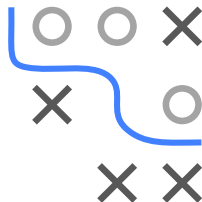
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- Here,
  - $\sum_{t=1}^T L(a_t, z_t)$  is the *cumulative loss of the learner*,
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# MEASURE OF QUALITY IN ONLINE LEARNING

- It seems natural to compare the incurred cumulative loss of the learner with the *best action(s) in hindsight*:

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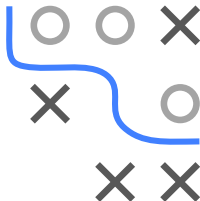
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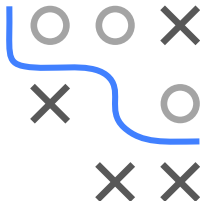


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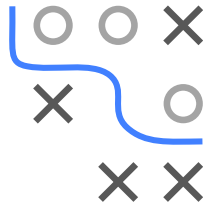
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- We refer to  $R_T$  as the (*cumulative*) *regret* of the online learner. It is easy to see that  $R_T = \sup_{a \in \mathcal{A}} R_T(a)$ .



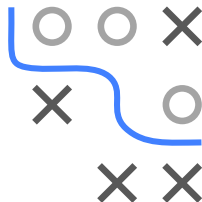
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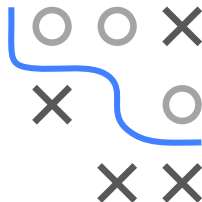
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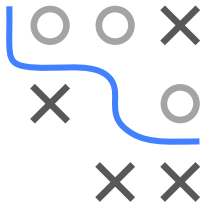
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- Formally, the following should hold

$$R_T = o(T).$$

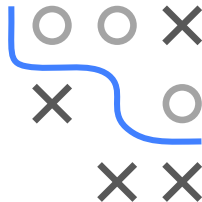
*Interpretation:* The average regret per time step (or per example) goes to zero:

$$\frac{1}{T} \left( \sum_{t=1}^T L(a_t, z_t) - \inf_{a \in \mathcal{A}} \sum_{t=1}^T L(a, z_t) \right) = \frac{R_T}{T} = o(1).$$



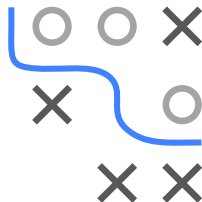
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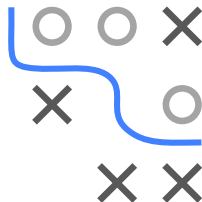
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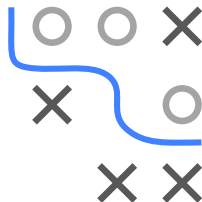
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- We will cover only the static regret in this lecture.

